ECE250: Signals & Systems Quiz-9

Date: 12th November, 2025 **Total Marks:** 10 **Duration:** 15 Minutes

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Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

[CO2, CO4] **[Q1]**: Consider the signal-

$$x(t) = e^{-5t}u(t - t_0) + e^{-\beta t}u(t)$$

(1) Compute the Laplace transform $X(s)$ and specify the region of convergence (ROC). **[7 Marks]**

(2) What are the constraints on the real and imaginary parts of β exist if the region of convergence of $X(s)$ is $\Re\{s\} > -4$? **[3 Marks]**